

AM 463: QUIZ 6
Oct. 24 in class
15 minutes: Closed Book
Answer all problems
Presentation will affect your grade

Question 1:

- i) Write down the shallow water equations in the presence of rotation.
- ii) Non-dimensionalize the equations and derive geostrophic flow. Define all terms (e.g. the Rossby number) you use.

Question 2:

In shallow water theory, the dispersion relation for Poincaré waves propagating in the x direction is given by

$$\sigma^2 = f^2 + gHk^2$$

- i) Find the phase and group velocities and discuss their behaviour in the long wave limit. Remind me what g and H are.
- ii) Discuss the behaviour in the short wave limit. Does rotation matter? Why or why not?